

Reducing Food Waste Through Portion-Controlled Bundling

*Designing smarter defaults that align what
people buy with what they actually eat*

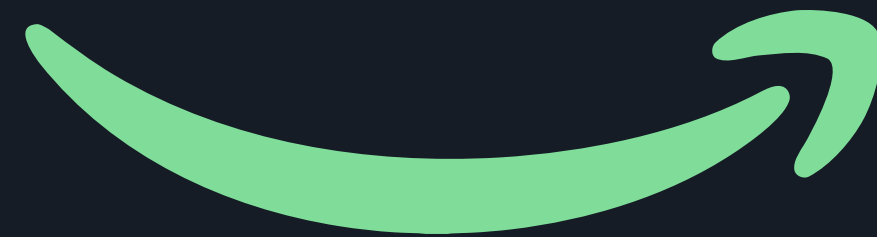
amazon fresh

Agenda

- [01] **Context**
- [02] **Research & Insights**
- [03] **Problem Statement**
- [04] **Solution**
- [05] **Impact & Outcomes**
- [06] **Next Steps**



Pick an existing digital ecosystem
and design a climate-positive
upgrade that fits inside it.



Why Household Food Waste?



[01]

Food loss and waste account for **8–10% of annual global GHG emissions.**



[02]

30-40% of the U.S. food supply is wasted, with households as the largest contributor.



[03]

U.S. households **waste over \$1,300+ annually** in uneaten food.



Key Demographics

A national study of ~1,000 U.S. households shows that food waste is behavioral and concentrated, making targeted intervention for high-waste households more effective than broad solutions.

Lowest waste

40%
Structured Planners

Deliberate shoppers with strong planning habits and active waste reduction

Moderate waste

47%
Flexible Planners

Casual, spontaneous shoppers with positive attitudes toward reducing waste

Highest waste

13%
Young Wasters

Younger, lower-income households with minimal planning habits who **waste food nearly 7× per week** (*compared to ~4.5× for other groups*), and waste significantly more **proteins, oils, and grains**

Key Insight

The smallest segment wastes the most food—especially expensive, high-impact items—making them the greatest opportunity for targeted intervention.

The Young Wasters would benefit most with **meal planning, food storage,** and **portion control.**

Why Amazon Fresh?



[01]

Aligned Users

70% of Amazon Fresh shoppers belong to the **BFY (Better-For-You) segment**, compared to 32% at the average U.S. grocery retailer.



[02]

System Scale

Amazon serves **5,000+ U.S. cities** with same-day grocery delivery and is shifting investment from physical stores to online grocery fulfillment.



[03]

Climate Commitment

Amazon's **net-zero by 2040** pledge and **\$2B Climate Pledge Fund** make household food-waste reduction a direct climate lever.



Where Waste Accumulates Today

At Cart

The cart experience optimizes for item accumulation rather than quantity accuracy at the household level, often leading to cart-overfilling.

It's easy to buy more than you need.



Delivery

Perishable items are delivered in fixed quantities that do not adapt to household consumption patterns.

Fresh food arrives faster than it can be used.



At Home

Variability in schedules and routines leads to a mismatch between purchase intent and actual consumption.

Schedules change, meals get skipped.



Leftovers

Post-delivery spoilage occurs outside the system, preventing learning or corrective feedback.

What goes uneaten goes unnoticed.



Amazon Tried to Solve This with Meal Kits... But Failed

- High food waste
- Low-margin
- Low customer retention
- High operational costs
- Convenience mismatch
- Strategic shift

Key Takeaway

This failure did not signal lack of demand. It revealed a rigid, high-cost model misaligned with real customer behavior.



r/wholefoods · 2y ago
[deleted]

What happened to the Amazon meal kits?

I see they are still available in some select Amazon Fresh stores, but I no longer see them in Whole Foods. They were supposed to be like Hello Fresh but better- no subscription! Hello!! Amazon Fesh!

0

5

NEW amazon meal kits

WE DO THE PREP. YOU BE THE CHEF.

No recipe hunting. No ingredient shopping. No tedious prep.

Making dinner doesn't need to be a chore. With Amazon Meal Kits, you can whip up a delicious, chef-designed dinner for two in about 30 minutes. We take on the boring stuff ahead of time, leaving just a few steps between you and an amazing dinner. Perfectly portioned ingredients, easy-to-follow recipes, great-tasting food. Simply delicious.

WE DO THIS...

CHEF-DESIGNED, EASY-TO-FOLLOW RECIPES

MOST OF THE PREP DONE IN ADVANCE!

PERFECTLY PORTIONED INGREDIENTS

YOU DO THIS!

From box to table in about 30 minutes

.....

r/amazon · 8y ago
harujin215

What happened to the Amazon meal kits?

I recently tried the Amazon meal kits in Amazon Fresh stores and they disappeared immediately. They became part of my weekly order for a few months, and then.. they disappeared... now there are other meal kits and other 3d party options which I tried and in some cases the amazon ones... has anyone noticed that as well as well as the amazon ones... may be a regional/temporal thing (NYC) or if they were just a one-time thing?

3

3



r/amazonfresh · 5y ago
[deleted]

When will they have, "Amazon Meal Kits"?

I understand they are not available due to when covid hit. Will they ever come back in the near future?

Yes they were over priced for what you got, but they were convenient and very tasty :(

As someone who doesn't know how to cook or takes a long time to cook raw ingredients, I liked it

5

5

Why Try Again And Why Now?



[01]

Market Size

Online grocery delivery is growing at **11%+ CAGR through 2030**, driven by convenience and digital adoption.



[02]

Consumer Behavioral Shift

The U.S. meal-kit market is **growing at ~10.7% annually**, signaling rising demand for digitally assisted meal decisions.



[03]

Amazon's Unique Leverage

Amazon is **scaling same-day grocery to 2,300+ U.S. cities**, consolidating physical formats and strengthening its end-to-end digital system.



Research & Insights

Competitive Analysis

Meal kits reduce choice but are rigid and waste-prone, while grocery delivery is flexible but drives over-purchasing and spoilage.



Opportunity Space

Combine flexibility with system-level quantity guidance to reduce household food waste at mass scale.

Key Insights

Concentrated

Household food waste is driven by a **small set of high-impact behaviors**, not by lack of awareness or motivation.



Invisible

Most food waste occurs **after delivery**, outside the system's visibility or control.

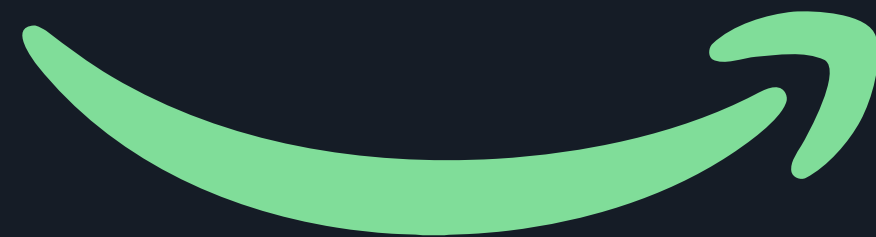


Structural

Existing solutions focus on products, not **system-level defaults** that shape quantities entering the home.



At the intersection of climate impact and digital grocery scale, **Amazon Fresh** has an opportunity to **reduce household**  **food waste** by shifting from optimizing what customers buy to **optimizing how much they actually eat.**



Solution

Smart Defaults Bundling

Smart Bundles dynamically group groceries and right-size quantities using behavioral signals to reduce food waste by default.

Click Here for Prototype

 **Your bundles help reduce unnecessary food waste**

Smart portions mean less goes to waste. We'll check in after delivery to keep improving your recommendations.

Estimated waste avoided

~0.8 lb

 **Recommended Bundles for You**

Designed to match how you usually cook and eat



Weeknight Chicken & Rice
Amazon Grocery Chicken Breast, 36 oz

Optimized for 1–2 dinners

\$23.94

View >



Salmon Power Bowl
Amazon Grocery Salmon Fillet, Amazon's Choice

Optimized for 1–2 dinners

\$20.97

View >

 **Portion-Controlled Bundle**

Quantities adjusted to reduce leftovers

3 items



Chicken Breast, Boneless

-

1

+



Jasmine Rice, Long Grain
2 lb bag
\$3.49

-

1

+

[Remove](#)

Bundle subtotal

\$14.97



ON123 

Delivered!

Today at 6:15PM

New York, NY









Your impact so far



3

Orders with bundles



2.1 lb

Est. waste avoided



2×

Feedback given

 **Your bundles are getting smarter**

Based on your recent feedback, we've adjusted your portion recommendations. Here's what changed:

 **We'll suggest slightly smaller portions next time**



Chicken Breast, Boneless Skinless
1 lb (2 breasts) → 0.75 lb (2 small breasts)



Broccoli Crowns, Organic
1 bunch (8 oz) → 1 bunch (6 oz)

Submit

This helps us personalize future recommendations

 **Quick feedback**

How were the portions?

For your "Weeknight Chicken & Rice"



ChickenBroccoliJasmine Rice

Did you finish most of these items?

 **Yes, just right**
The portions worked perfectly

 **Mostly**
A little was left over

 **No, too much**
I had significant leftovers

Submit

This helps us personalize future recommendations

 **Often bought as part of a meal bundle**

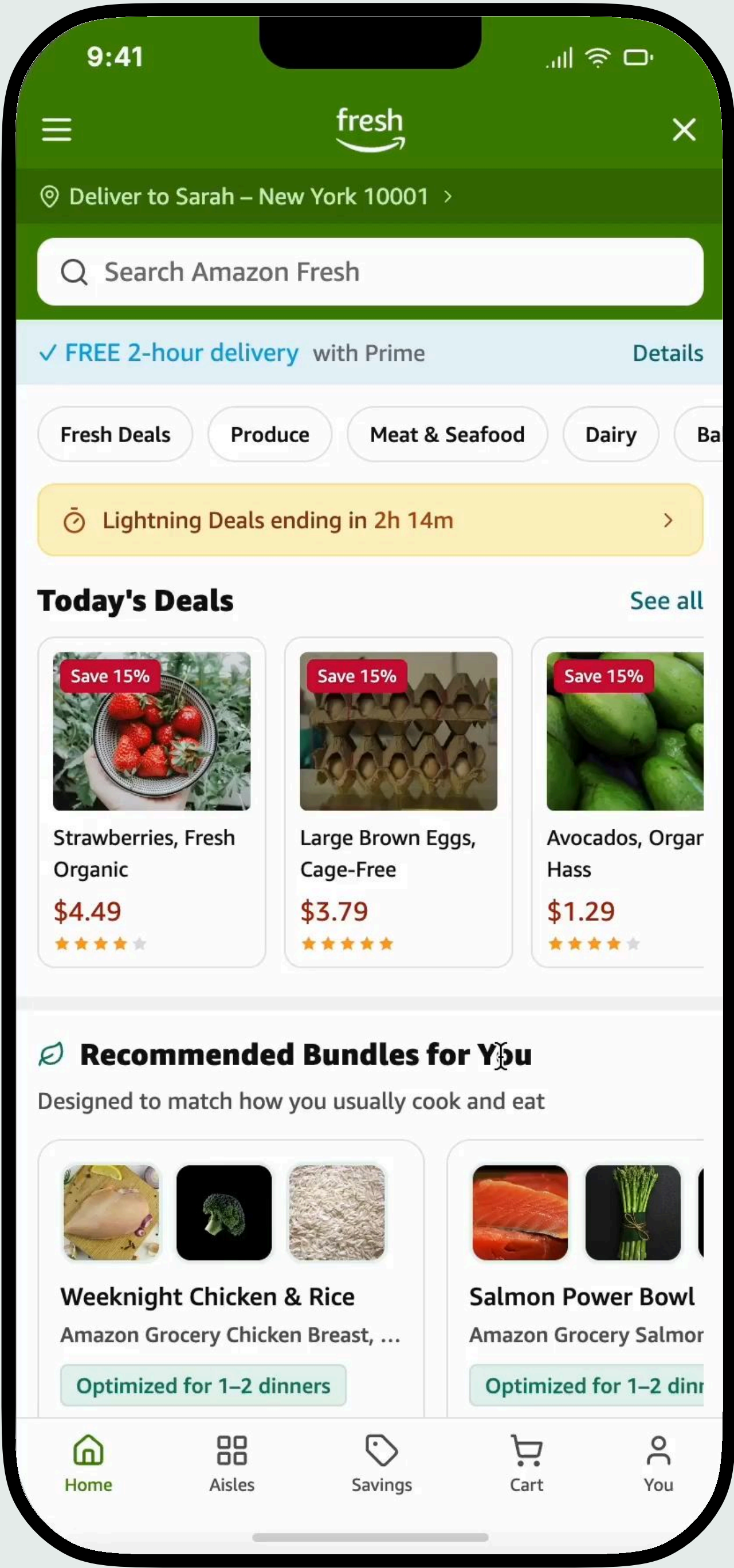
Choose your quantity

Solution

Touchpoint 1: Home, Bundles, Product Pages

Smart Bundles surface personalized, right-sized groupings across home, bundle, and product pages, making quantity guidance visible by default while preserving full user flexibility.

Click Here for Prototype

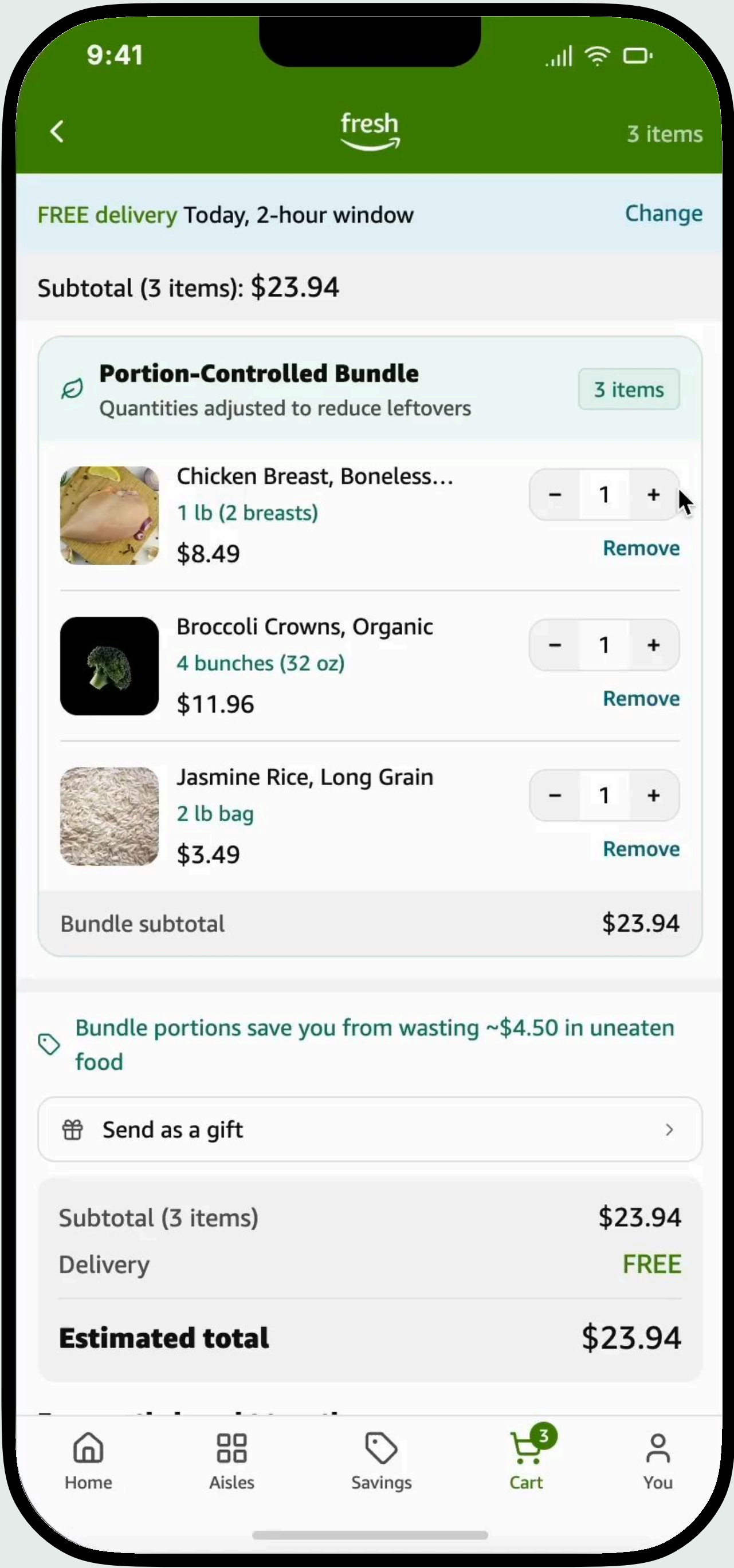


Solution

Touchpoint 2: Cart, Checkout, Confirmation

Bundles stay grouped as you check out, let you easily adjust quantities, and show how choosing smarter portions helps reduce food waste, all while the system keeps learning for next time.

[Click Here for Prototype](#)

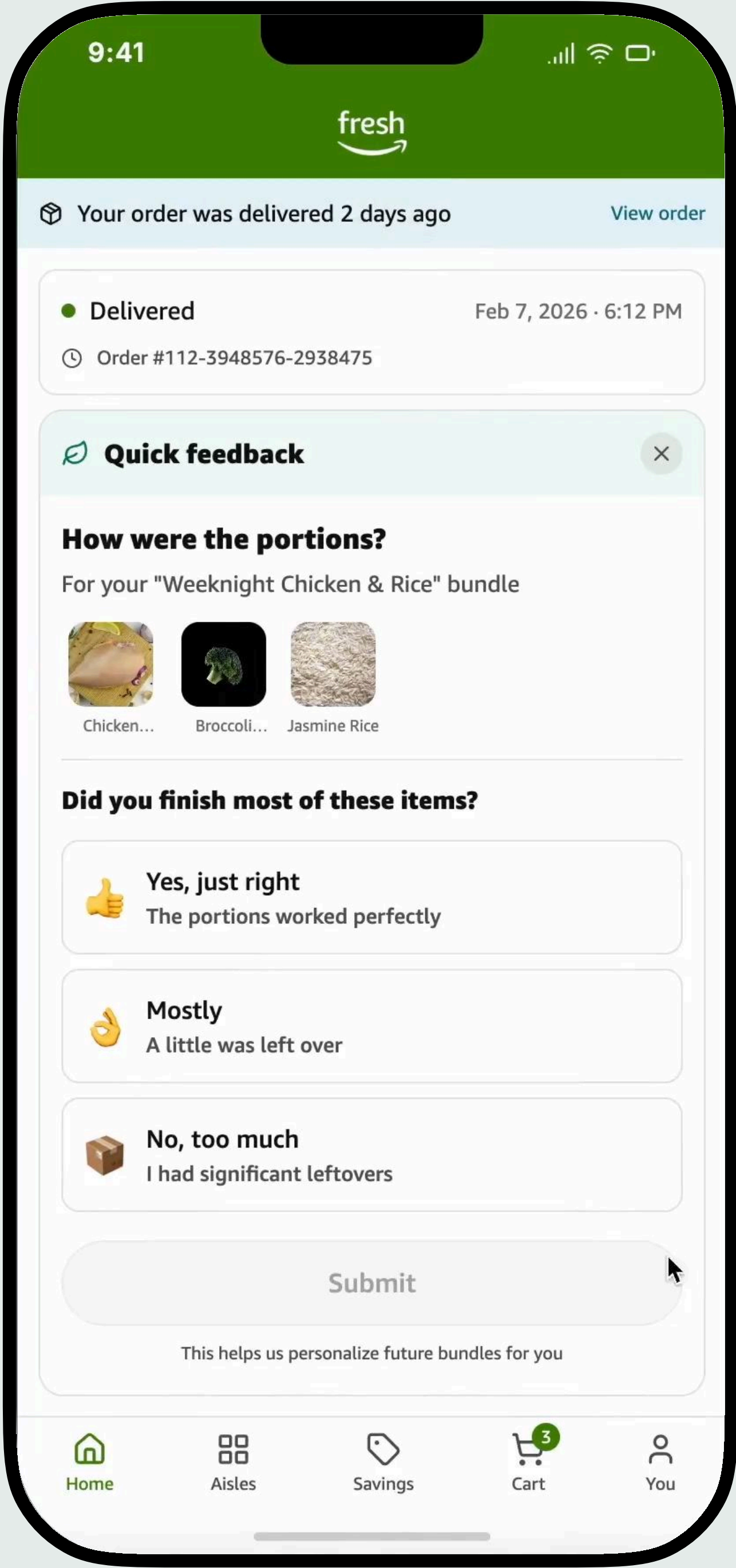


Solution

Touchpoint 3: Feedback, Learning

After delivery, users can share quick, optional feedback about portion sizes, and the system clearly shows how that input makes future bundles smarter so that quantities improve over time without any extra effort.

[Click Here for Prototype](#)



Solution

Touchpoint 4: Delivery

At delivery, users receive right-sized portions in their bundles, reducing leftovers at home without requiring any change to how they cook or eat.

[Click Here for Prototype](#)

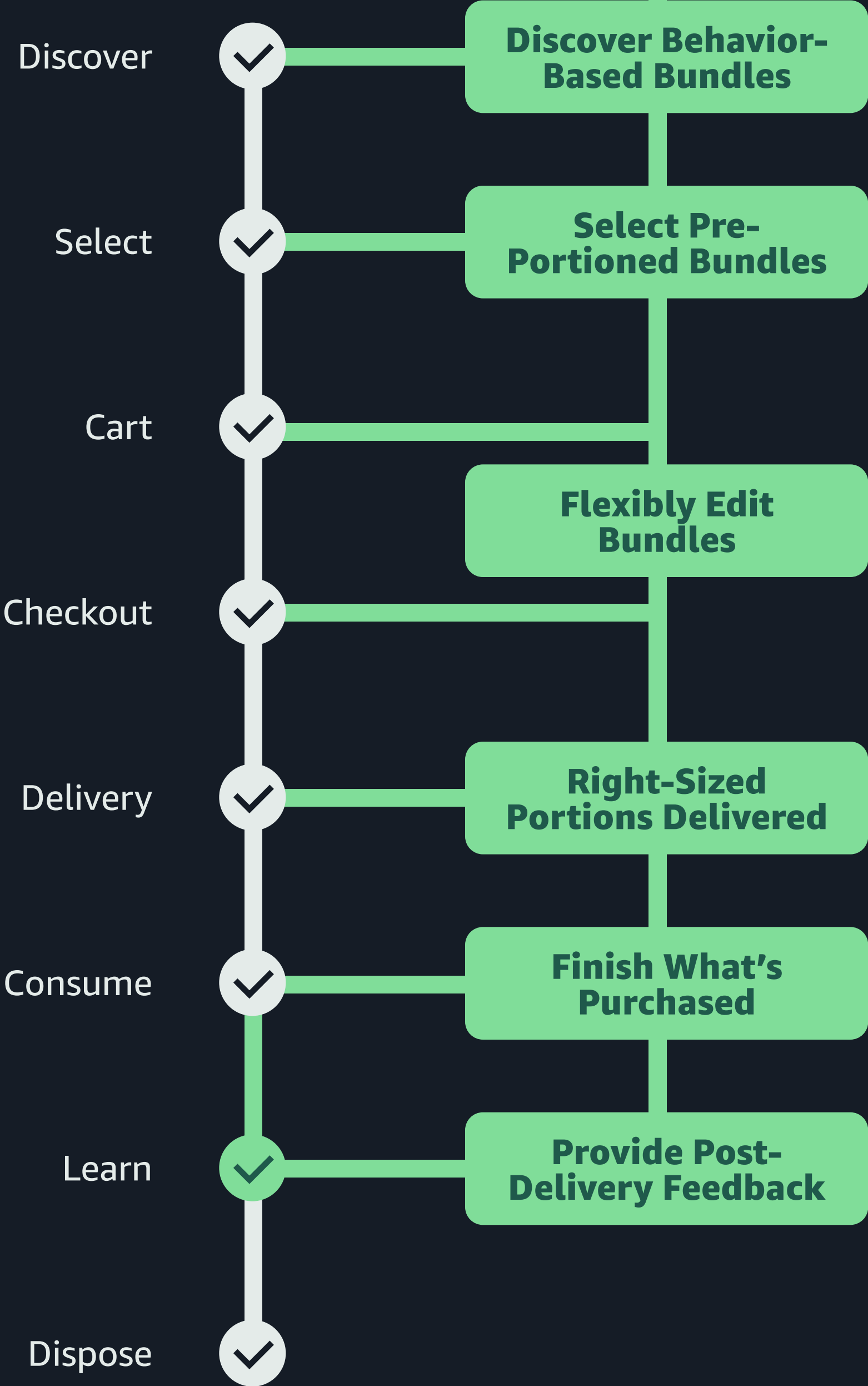


Impact & Outcomes

Ecosystem Map

Smart Bundles create a seamless closed loop across the entire system, using behavior data to continuously refine portions and reduce waste without changing how users shop or cook.

- Existing Ecosystem
- Where Bundles Intervene



Impact

Systems

Instead of treating grocery shopping as a one-time transaction, the system learns from what actually gets eaten and **continuously adjusts future defaults.**



User

Users now **waste less food** because the system helps them **buy the right portions by default** without added planning or effort.



Climate

Less food ends up in household trash, **reducing emissions** from producing, transporting, and disposing of food that was never eaten.



Business

Smarter defaults reduce spoilage and returns while **increasing customer trust, retention, and long-term value at scale.**



Impact & Outcomes

Impact Metrics

User

≥20% adoption

of portion-controlled bundles with no increase in checkout abandonment within **3 months**.



Climate

15-20% reduction

in post-delivery spoilage signals for bundled categories within **12 months**.



Business

10-15% lift

in reorder consistency and repeat purchase rate among bundled users within **6 months**.



Next Steps



[01]

Optimize

Validate bundling reduces over-purchasing through targeted user testing, then **refine Smart Defaults using real-world usage data.**

[02]

Expand

Scale portion intelligence across Fresh, Whole Foods, and Subscribe & Save as an embedded default.

[03]

Measure

Track long-term waste reduction, habit formation, and measurable sustainability impact.

Thank You!

